



5213 - SCUBADive: A JWST Archival Analysis of the Most Dust-Obscured Galaxies Across Cosmic Time

Cycle: 3, Proposal Category: AR

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Jed McKinney (PI)	University of Texas at Austin
Prof. Caitlin M. Casey (CoI)	University of Texas at Austin
Dr. Lu Shen (CoI)	Texas A & M University
Arianna Long (CoI)	University of Texas at Austin
Olivia Cooper (CoI)	University of Texas at Austin
Dr. Maximilien Franco (CoI)	University of Texas at Austin
Dr. Sinclair Manning (CoI)	University of Massachusetts - Amherst
Hollis Akins (CoI)	University of Texas at Austin
Dr. Margherita Talia (CoI) (ESA Member)	Universita di Bologna
Fabrizio Gentile (CoI) (ESA Member)	Universita di Bologna
Erini Lambrides (CoI)	NASA Goddard Space Flight Center
Elaine Gammon (CoI)	University of Georgia Research Foundation, Inc.

OBSERVATIONS

ABSTRACT

What are the optical and near-infrared counterparts to sub-millimeter galaxies (SMGs) and what is their redshift distribution? Are they overwhelmingly disturbed by galaxy-galaxy interactions, or can the high star-formation rates commonly found in SMGs reside within smooth disks? Historically, the answers to these questions have been limited to lower redshift ($z < 1-2$) sub-mm sources where ground- and space-based telescopes can still detect the rest-optical light from stars. Now, JWST is sensitive enough to detect the optical/near-IR counterparts to SMGs up to much higher

redshifts ($z>2-4$) - enabling a new perspective on a pivotal category of galaxies which dominates the star-formation rate budget for most of cosmic time. In this program we propose an Archival analysis of SMGs to uncover their JWST counterparts, leveraging JWST's ability to detect stellar light through extreme optical attenuation and in distant galaxies. We will use the combined optical/near-IR (JWST) through far-infrared/sub-mm (ALMA, SCUBA2) data to (1) measure the full redshift distribution of SMGs down to fainter sub-mm fluxes than ever explored before, (2) measure the volume density of the elusive population of $z>4$ SMGs, and (3) spatially resolve the stellar properties of SMGs to distinguish principle quenching pathways. To answer these questions we will identify the JWST counterparts to the 706 SMGs that fall within the NIRCам footprint of COSMOS-Web (Cycle 1). This archival program will produce the largest volume-limited SMG sample with multi-wavelength SEDs that include JWST data, and also pave the way for a similar analysis over other JWST+ALMA fields.

OBSERVING DESCRIPTION

This is an archival proposal.